Coverage Report for SafetyFunction

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Analysis Information

Coverage Data Information

Collected in version	(R2023b)

Model Information

Model version	5.13
Author	DAUIN-Win10
Last saved	Mon Jun 24 18:08:04 2024

Harness information

Harness model(s)	SafetyFunction
Harness model owner	controller

Simulation Optimization Options

Default parameter behavior	tunable
Block reduction	forced off
Conditional branch optimization	on

Coverage Options

Analyzed model	SafetyFunction/SaftyFunctions
Logic block short circuiting	off
MCDC mode	masking

Tests

Test	Started execution	Ended execution
Run 29	24-Jun-2024 18:08:20	24-Jun-2024 18:09:08

Summary

Model Hierarchy/Complexity **Run 29**

		Decision	Condition	MCDC	Execution
1. SaftyFunctions	6	100%	88%	75%	NA
2. . . . SF: SaftyFunctions	5	100%	88%	75%	NA

Details

1. SubSystem block "SaftyFunctions"

Child Systems: SaftyFunctions

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	6
Decision	NA	100% (6/6) decision outcomes
Condition	NA	88% (7/8) condition outcomes
MCDC	NA	75% (3/4) conditions reversed the outcome
Execution	NA	NA

2. Chart "SaftyFunctions"

Justify or Exclude

Parent: SafetyFunction/SaftyFunctions

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	5
Condition	NA	88% (7/8) condition outcomes
Decision	100% (2/2) decision outcomes	100% (6/6) decision outcomes
MCDC	NA	75% (3/4) conditions reversed the outcome

Decisions analyzed

Substate executed	100%
State "SafeStateOFF"	999/8000
State "StateStateON"	7001/8000

Transition "[abs(e)<=0.02&&CANdisconnection==0]" from "StateStateON" to "SafeStateOFF"

Justify or Exclude

Parent:	SafetyFunction/SaftyFunctions

Metric	**Coverage**
Cyclomatic Complexity	2
Condition	100% (4/4) condition outcomes
Decision	100% (2/2) decision outcomes
MCDC	100% (2/2) conditions reversed the outcome

```
1  [abs(e)<=0.02&&CANdisconnection==0]
```

#1: [abs(e)<=0.02&&CANdisconnection==0]

Decisions analyzed

abs(e)<=0.02&&CANdisconnection==0	100%
false	6500/7001
true	501/7001

Conditions analyzed

Description	**True**	**False**
abs(e)<=0.02	2001	5000
CANdisconnection==0	501	1500

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	**True Out**	**False Out**
abs(e)<=0.02&&CANdisconnection==0		
abs(e)<=0.02	**T**T	**F**x
CANdisconnection==0	T**T**	T**F**

Transition "[abs(e)>=0.02||CANdisconnection==1]" from "SafeStateOFF" to "StateStateON"

Justify or Exclude

Parent:	SafetyFunction/SaftyFunctions

Metric	**Coverage**
Cyclomatic Complexity	2

Condition	75% (3/4) condition outcomes
Decision	100% (2/2) decision outcomes
MCDC	50% (1/2) conditions reversed the outcome

```
1  [abs(e)>=0.02||CANdisconnection==1]
```

#1: [abs(e)>=0.02||CANdisconnection==1]

Decisions analyzed

abs(e)>=0.02||CANdisconnection==1	100%
false	498/999
true	501/999

Conditions analyzed

Description	True	False
abs(e)>=0.02	501	498
CANdisconnection==1	0	498

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	True Out	False Out
abs(e)>=0.02||CANdisconnection==1		
abs(e)>=0.02	Tx	FF
CANdisconnection==1	(FT)	FF